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JAVA PROJECT REPORT ON

The Restaurant Order Processing and Billing System

Submitted by

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DECLARATION

I hereby declare that the project entitled "**The Restaurant Order Processing and Billing System″** submitted in partial fulfillment of the requirements for the course **JAVA PROGRAMMING LANGUAGE** is a **record of my original work** carried out under the guidance of **Dr.BOOPALAN SUNDARAMOORTHY,**, Technical Head, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

Ihave acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

HARIRAM K

CERTIFICATE

This is to certify that the project entitled **"The Restaurant Order Processing and Billing System"** is a **bona fide work** carried out by HARIRAM K,- **'A'(2025 - 2028)**, **Department of Computer Science**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **JAVA PROGRAMMING LANGUAGE.**

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: Coimbatore

Date:10/04/2026

Submitted for the Viva-Voce Examination held on **10/04/2026**

Guide:
(Technical Head, KGCAS)

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Head of the Department
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TABLE OF CONTENTS

S no	Topics	Page no
01	Introduction	01
1.1	Objective	01
1.2	Overview	02
1.3	Features	03
2.0	Program analysis	04
2.1	Problem defination	05
2.2	Project overview	05
2.3	Scopes and Limitation	06
3.0	System design	07
3.1	Modules	08
4.0	Methodology	09
4.1	Problem analysis	09
4.2	System design	09
4.3	Program development	10
4.4	Testing and validation	10
4.5	Result and documentation	11
4.6	Algorithm	11
4.7	Flowchart	12
5 .0	Source code	13
6.0	Input and output screen	14-15
7.0	Discussion and conclusion	16
7.1	Discussion	16
7.2	Conclusion	16
8.0	References	17

ABSTRACT

The Restaurant Order Processing and Billing System is a software application designed to automate the day-to-day operations of a restaurant. The main objective of this system is to simplify order management, improve billing accuracy, and enhance overall customer service. Traditionally, restaurants rely on manual methods for taking orders and generating bills, which can lead to errors, delays, and inefficiencies. This system addresses those issues by providing a computerized solution where orders can be placed, updated, and processed in real time. It allows staff to select menu items, calculate totals automatically, and generate bills instantly. The system typically includes features such as menu management, order tracking, bill generation, and basic reporting. It reduces human effort, minimizes calculation mistakes, and speeds up service delivery. Additionally, it helps in maintaining records of sales and transactions, which can be useful for business analysis. Overall, the Restaurant Order Processing and Billing System improves operational efficiency, ensures accuracy, and provides a better experience for both staff and customers.

1. INTRODUCTION

The Restaurant Order Processing and Billing System is a computer-based application developed to manage and streamline restaurant operations efficiently. In today's fast-paced environment, restaurants require quick and accurate service to satisfy customer needs. Manual methods of taking orders and preparing bills are time-consuming and prone to human errors, which can affect service quality and customer satisfaction. This system is designed to automate the process of order taking, processing, and billing. It enables restaurant staff to record customer orders digitally, calculate the total cost automatically, and generate accurate bills in a short time. By replacing traditional paper-based methods, the system improves speed, reduces mistakes, and enhances overall productivity. The application also helps in organizing menu items, tracking orders, and maintaining transaction records. It ensures better coordination between different sections of the restaurant, such as kitchen and billing counters. As a result, the system not only improves operational efficiency but also provides a smooth and convenient experience for customers. In conclusion, the Restaurant Order Processing and Billing System is an effective solution that supports modern restaurant management by making operations faster, easier, and more reliable.

1.1 Objective

- The mainobjectives of the Restaurant Order Processing and Billing System are:

- To automate the process of taking and managing cus tomer orders.
- To generate accurate bills quickly and reduce manual calculation errors.
- To improve the speed and efficiency of restaurant operations.
- To maintain proper records of orders and transactions.
- To enhance customer satisfaction by providing faster service.
- To reduce paperwork and human effort in daily activities.

1.2 Overview

The Restaurant Order Processing and Billing System is a software application designed to handle restaurant activities such as order taking, processing, and billing in an efficient manner. In this system, restaurant staff can select items from a digital menu, place orders, and automatically calculate the total bill. The system keeps track of all orders and

generates bills instantly. It also stores transaction details for future reference.

This application replaces traditional manual methods, ensuring better accuracy, faster service, and improved coordination between different sections like the kitchen and billing counter. Overall, it simplifies restaurant management and enhances productivity.

1.3 Features

Key features of the system include:

- User-friendly Interface – Easy to use for staff with minimal training.
- Menu Management – Add, update, or delete food items and prices.
- Order Processing – Quick and accurate order entry and m odification.
- Automatic Billing – Calculates total amount, taxes, and generates bills instantly.
- Order Tracking – Keeps track of pending and completed orders.
- Data Storage – Stores order history and transaction details.
- Error Reduction – Minimizes mistakes in billing and order handling.
- Time Saving – Speeds up service and reduces waiting time for customers.

2. PROGRAM ANALYSIS

Program analysis focuses on understanding the existing system, identifying problems, and defining requirements for the new system. In traditional restaurants, order processing and billing are done manually, which leads to errors, delays, and poor record management. There is difficulty in tracking accurately, and maintaining transaction records. The proposed system analyzes these issues and provides a computerized solution to improve speed, accuracy, and efficiency. It identifies user needs such as easy order entry, automatic billing, and proper data storage.

2.1 Problem Definition

- The main pr oblem is the inefficiency of manual order processing and billing systems in restaurants.
- Key problems include:
- Time-consuming order taking and bill preparation
- Errors in manual calculations
- Difficulty in managing multiple orders
- Lack of proper record maintenance
- Poor coordination between staff and kitchen
- Increased customer waiting time
- The system is designed to solve these problems by automating the entire process.

2.2 Project Overview

TheRestaurant Order Processing and Billing System is a software application that automates restaurant operations.

- The system allows staff to: Take customer orders digitally
- Select items from a predefined menu Automatically calculate
- the total bill Generate and print bills Store transaction details
- for future use It improves communication between order-taking
- staff and the kitchen while ensuring fast and accurate service.
- The system is suitable for small to medium-sized restaurants.

2.3 Scope and Limitations

Scope

- Can be used in restaurants, cafes, and food outlets
- Handles order processing and billing efficiently
- Reduces manual work and improves accuracy
- Maintains digital records of transactions
- Can be enhanced with features like online ordering and payment integration

Limitations

- Requires basic computer knowledge to operate
- Initial setup cost for hardware and software
- May not support very large restaurant chains without upgrades
- Dependent on system reliability (power/system failures can affect operations)
- Limited features if not integrated with advanced technologies

3. System Design

- System design defines the structure and working of the Restaurant Order Processing and Billing System. It describes how different components (modules) of the system interact to perform tasks efficiently.
- The system is designed using a modular approach, where each module handles a specific function. This improves flexibility, maintainability, and ease of use.

3.1 Modules

1.User Interface Module

- Provides interaction between user and system
- Allows staff to input orders and view menu
- Simple and user-friendly design

2. Menu Management Module

- Stores details of food items and prices
- Allows adding, updating, and deleting menu items
- Displays menu for order selection

3. Order Processing Module

- Handles customer order entry
- Allows modification or cancellation of orders
- Sends order details to the kitchen (logically)

4. Billing Module

- Calculates total amount automatically
- Includes taxes and discounts (if any)
- Generates and prints the bill

5. Database Module

- Stores customer orders and transaction records
- Maintains menu details and billing history
- Ensures data is organized and easily retrievable

6. Reporting Module (Optional)

- Generates daily or monthly sales reports
- Helps in analyzing business performance
- Useful for decision-making

4. Methodology

- The methodology describes the step-by-step process followed to develop the system. It ensures proper planning, design, development, and implementation.

4.1 Problem Analysis

- Study the existing manual system used in restaurants
- Identify issues like delays, billing errors, and poor record keeping
- Understand user requirements (fast billing, easy order entry, data storage)
- Define system goals to improve efficiency and accuracy

4.2 System Design

- Divide the system into modules (UI, Menu, Order, Billing, Database)
- Design input screens for order entry and menu display
- Plan database structure to store items and transactions
- Define system workflow from order taking to bill generation
- Ensure user-friendly and efficient design

4.3 Program Development

- Choose programming language (e.g., Java)
- Develop each module step by step:
- Menu management
- Order processing
- Billing system
- Integrate all modules into a complete system
- Implement logic for calculations and data storage
- Ensure code is simple, efficient, and error-free

4.4 Testing and Validation

- Test each module individually (Unit Testing)
- Test the complete system (System Testing)
- Check for:
- Correct bill calculation
- Proper order processing
- Data storage accuracy
- Fix errors and bugs
- Validate system performance with sample data

4.5 Result and Documentation

- The system successfully automates order processing and billing
- Reduces manual errors and saves time
- Provides accurate and fast billing
- Documentation includes:
- System description
- User manual
- Program code explanation
- Helps users understand and operate the system easily

Algorithm

Step-by-Step Algorithm

1. Start
2. Display menu items
3. Accept user choice
4. Enter quantity
5. Calculate item total = price × quantity
6. Add to overall total
7. Ask for next order (Yes/No)
8. If Yes, repeat from step 2
9. If No, generate total bill
10. Display bill
11. Stop

Flowchart

Start
↓
Display Menu
↓
Enter Choice
↓
Enter Quantity
↓
Calculate Total
↓
Add to Bill
↓
More Orders? → Yes → Repeat
↓ No
Generate Bill
↓
Display Bill
↓
End

Source code

```
import java.util.Scanner;

public class RestaurantBilling {
public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

      String[] items = {"Burger", "Pizza", "Pasta", "Sandwich"};
      int[] prices = {100, 200, 150, 80};

      int choice, quantity;
      int total = 0;
      char more;

      do {
         System.out.println("\n--- MENU ---");
         for (int i = 0; i < items.length; i++) {
            System.out.println((i + 1) + ". " + items[i] + " - Rs." + prices[i]);
         }

         System.out.print("Enter your choice: ");
         choice = sc.nextInt();

         if (choice >= 1 && choice <= items.length) {
            System.out.print("Enter quantity: ");
            quantity = sc.nextInt();

            int cost = prices[choice - 1] * quantity;
            total += cost;

            System.out.println("Added: " + items[choice - 1] + " x " + quantity);
         } else {
            System.out.println("Invalid choice!");
         }

         System.out.print("Do you want to order more? (y/n): ");
         more = sc.next().charAt(0);

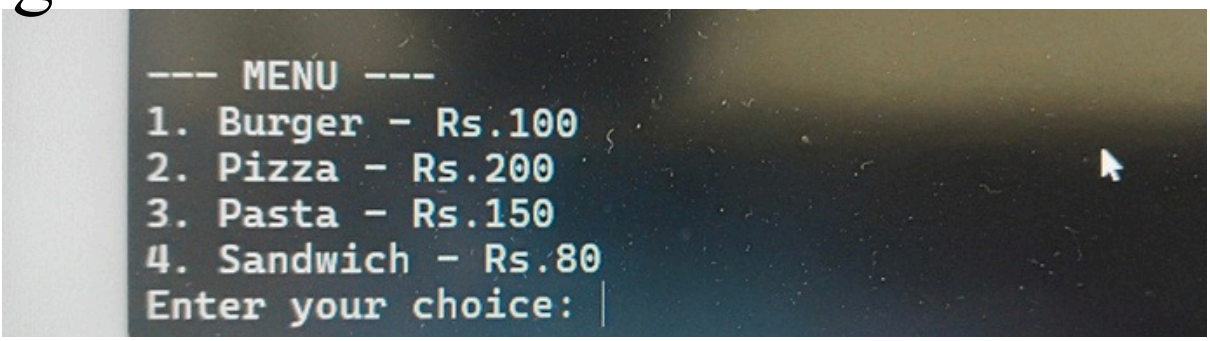
      } while (more == 'y' || more == 'Y');

      System.out.println("\n--- FINAL BILL ---");
      System.out.println("Total Amount: Rs." + total);
      System.out.println("Thank you!");

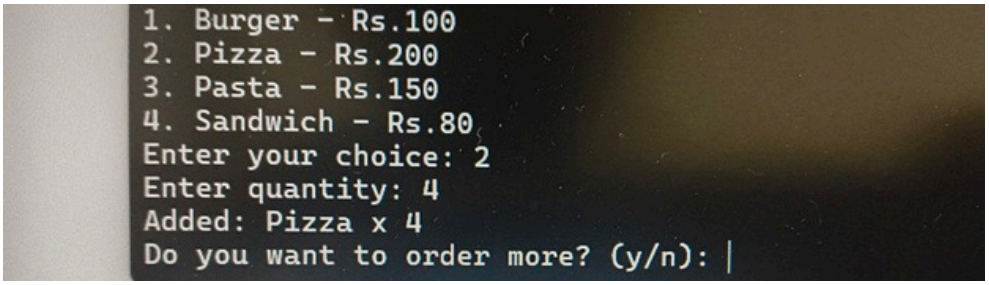
      sc.close();
   }
```

6.INPUT AND OUTPUT SCREENS

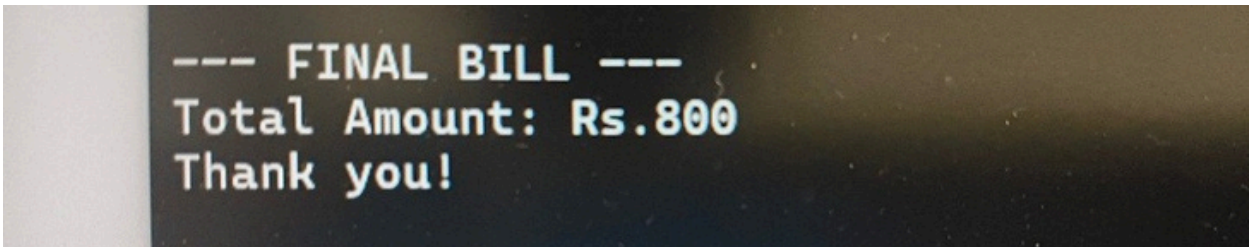
Login



Main



Output



Each screenshot should be labeled.

Example:

Figure 1: Menu Screen

Figure 2: Add to cart screen

Figure 3: Bill screen

7. DISCUSSION & CONCLUSION

7.1 Discussion

- This project demonstrates:
- Basic Java programming
- Real-world application
- Use of loops and arrays
- It can be improved by:
- Adding GUI (Swing/JavaFX)
- Connecting database
- Adding payment options

7.2 Conclusion

- The Restaurant Order Processing and Billing System successfully automates order handling and billing. It is simple, efficient, and use ful for beginners learning Java.

8. REFERENCES

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